

AQUADILLA HERNANDEZ INTL, Puerto Rico

WMO: 785140

Lat: 18.498N

Lon: 67.129W

Elev: 218

StdP: 14.58

Time Zone: -4.00 (W04)

Period: 94-17

WBAN: 11603

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	67.6	68.1	59.4	76.3	72.7	61.5	82.4	72.6	28.5	81.1	24.8	79.4	5.1	120	0.394

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.3	89.9	78.6	89.3	78.6	87.9	78.6	80.7	86.5	79.8	86.0	79.2	85.5	15.0	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.0	151.6	85.5	78.1	147.2	85.0	77.3	143.1	84.5	44.4	86.8	43.6	86.4	43.0	86.0	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.4	22.7	21.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	78.8	75.9	75.6	76.0	77.8	78.9	80.3	81.0	81.7	81.1	80.5	79.0	77.1
	DBStd	2.68	1.92	1.57	1.53	1.60	1.64	1.34	1.59	1.71	1.50	1.54	1.87	1.79
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10498	802	716	807	833	896	909	962	981	934	947	870	841
	CDD65	5023	337	296	342	383	431	459	497	516	484	482	420	376
	CDH74	41102	1882	1698	2164	2875	3482	4261	4903	5190	4682	4254	3265	2446
	CDH80	11113	297	266	378	639	864	1209	1527	1763	1515	1312	846	497
Wind	WSAvg	10.0	11.1	11.1	11.2	11.2	9.6	10.0	11.2	10.1	8.0	7.5	9.2	10.4
Precipitation	PrecAvg	61.5	2.8	3.1	3.7	5.5	7.6	6.6	4.5	5.7	5.8	6.5	6.0	4.7
	PrecMax	83.1	5.7	7.5	11.3	15.1	19.6	13.5	9.8	10.5	11.1	14.8	15.2	10.7
	PrecMin	42.0	0.1	0.2	0.7	0.7	1.9	2.2	1.5	1.5	2.1	2.0	1.7	0.9
	PrecStd	10.7	1.5	2.0	2.4	4.1	4.0	2.7	2.1	2.4	2.7	3.1	3.5	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.3	87.8	86.0	87.6	88.1	89.1	90.0	91.7	91.1	91.4	91.3	89.2
		MCWB	71.5	76.1	75.9	76.6	77.8	79.4	78.2	79.6	78.7	78.8	80.4	72.9
	2%	DB	84.4	84.2	84.4	86.0	86.7	87.7	89.3	89.8	89.2	88.2	87.9	85.9
		MCWB	75.7	74.5	75.1	76.3	77.6	78.5	78.1	78.9	79.0	79.3	78.8	77.6
	5%	DB	82.7	82.7	83.5	84.6	85.8	86.3	87.8	88.1	87.8	86.9	86.0	84.2
		MCWB	74.7	74.1	74.6	75.3	77.3	78.2	78.1	78.9	78.8	78.8	78.0	76.1
	10%	DB	81.3	81.5	82.3	84.0	84.4	85.7	86.3	87.4	86.3	86.0	84.5	82.7
		MCWB	73.7	73.6	74.0	74.9	76.6	78.0	78.2	78.8	78.3	78.3	76.9	75.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.5	78.3	78.7	78.8	79.9	80.6	80.9	81.4	81.3	81.1	79.8	79.0
		MCDB	83.0	82.3	82.8	84.9	85.7	86.1	86.1	87.0	87.3	86.4	85.7	84.1
	2%	WB	77.0	76.8	77.2	77.4	78.8	79.2	79.4	80.5	79.8	80.1	78.9	77.8
		MCDB	82.1	81.7	82.3	84.0	84.9	85.4	85.7	86.9	86.6	85.8	85.0	83.4
	5%	WB	75.4	75.4	76.2	76.6	78.0	78.7	79.0	79.4	79.1	79.1	78.2	76.7
		MCDB	80.9	80.9	81.8	83.2	84.1	85.1	85.6	85.9	86.0	85.2	84.2	82.7
	10%	WB	74.3	74.4	74.9	75.6	77.2	78.1	78.4	78.9	78.5	78.6	77.4	75.6
		MCDB	79.9	80.2	81.1	82.4	83.3	84.3	85.1	85.5	85.5	84.7	83.3	81.4
Mean Daily Temperature Range	5% DB	MDBR	12.0	12.9	13.1	12.2	11.7	11.5	11.6	12.3	11.9	12.4	11.8	12.4
		MCDBR	14.2	14.6	14.5	13.4	13.3	12.3	13.2	13.5	13.4	13.8	13.8	14.1
		MCWBR	8.5	8.7	8.6	7.5	7.6	7.3	6.9	6.8	7.3	7.8	7.2	8.2
	5% WB	MCDBR	12.7	13.6	13.9	12.6	12.2	11.7	11.5	12.1	11.8	11.9	11.7	12.5
		MCWBR	9.1	10.0	10.2	8.1	7.9	7.5	7.1	7.5	7.2	7.8	7.3	8.6
Clear-Sky Solar Irradiance	taub	0.365	0.384	0.382	0.412	0.450	0.524	0.518	0.485	0.435	0.386	0.383	0.374	
	taud	2.574	2.510	2.514	2.449	2.363	2.112	2.128	2.252	2.428	2.578	2.585	2.580	
	Ebn at Noon	292	291	294	284	270	248	250	260	274	287	284	285	
	Edh at Noon	30	34	34	37	40	51	50	44	37	31	30	29	
All-Sky Solar Radiation	RadAvg	1465	1703	1893	1989	1916	1941	1967	1928	1814	1643	1431	1382	
	RadStd	56	35	76	104	127	97	86	61	128	104	107	62	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	+0.59	+1.06	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+203	+202

Nomenclature: See separate page